Politecnico di Torino

Master's Degree in Mathematical Engineering

Material for Thesis

c– Data Pre-processing Pipeline
Intra and Inter Analysis

Elisabetta Roviera s328422

Contents

Abstract

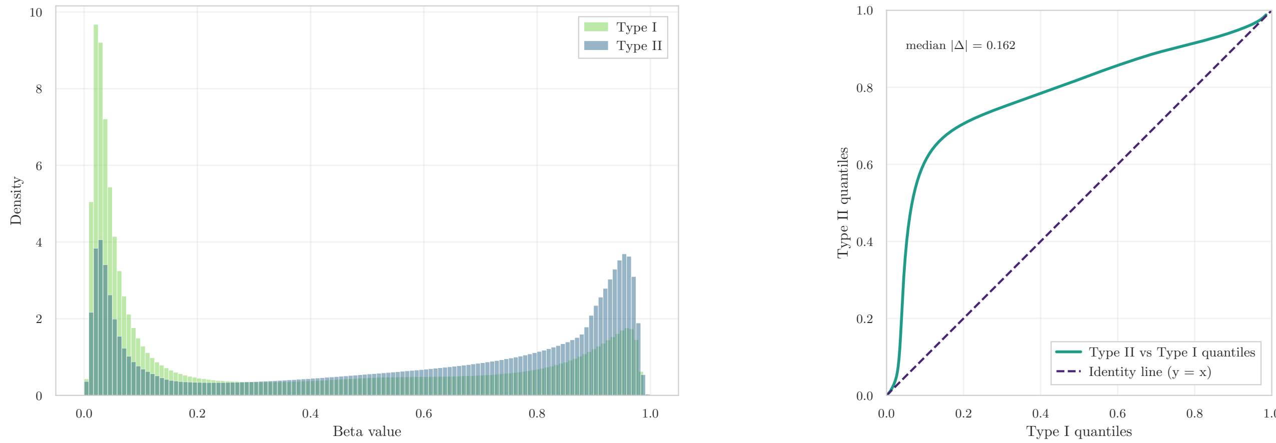
This document describes the data pre-processing pipeline applied to the three breast-tissue DNA methylation datasets employed in this thesis (GSE69914, GSE225845, GSE287331). The pipeline follows a unified methodological framework while accounting for dataset-specific technical characteristics. Core steps include data ingestion and alignment, missing-value handling, probe-level technical filtering, probe-design bias diagnostics, β-to-M value transformation, and the generation of harmonized outputs for subsequent analyses. Special attention is devoted to identifying and correcting array-specific artifacts in order to ensure comparability across cohorts.

All procedures are implemented in Jupyter notebooks, available in the thesis GitHub repository: 02-data-pre-processing-gse69914.ipynb, 02-data-pre-processing-gse225845.ipynb, 02-data-pre-processing-gse287331.ipynb, and 02-inter-analysis-data-pre-processing.ipynb.

Together, these notebooks define the final pre-processing stage of the analytical workflow, providing standardized and reproducible inputs for all subsequent modeling and biological interpretation steps.

1 Intra-Dataset Pre-processing

Data pre-processing represents a foundational stage in DNA methylation analysis, particularly in cancer epigenomics, where technical artefacts, probe-design biases, and unwanted sources of variation can obscure or distort genuine biological signals. Before performing feature selection, or machine-learning models, it is essential to ensure that the methylation matrices used for statistical modelling are accurate, and free from unreliable probes or batch-associated effects.



(a) β-value distribution by Infinium probe design (Type I vs. Type II). The near-overlapping shapes indicate effective correction of probe-type bias.

(b) Q–Q plot comparing quantiles of Type II vs. Type I probes. The alignment to the diagonal ($y = x$) confirms balanced signal distributions after normalization.

Figure 1: Diagnostic evaluation of Infinium Type I/II probe bias correction in the GSE69914 dataset. The consistent β-value and quantile distributions demonstrate the effectiveness of the BMIQ normalization previously applied.

In the context of breast cancer, where early epigenetic alterations may be subtle and highly localized, careful preprocessing is crucial to avoid confounding Normal–Adjacent differences with technical noise. Several well-established sources of unreliability—cross-reactive probes, SNP-affected loci, non-CpG targets, Infinium Type I/II probe-design bias, and batch effects—must be addressed through a systematic and reproducible workflow.

The primary goal of this section is therefore to construct, for each dataset, a high-quality and biologically coherent methylation matrix suitable for downstream analyses. To this end, all datasets undergo the same structured pipeline: initial alignment and quality checks; removal of unreliable CpGs through literature-based technical filtering; and conversion from β-values to M-values for statistical modelling.

These steps ensure methodological consistency across the three cohorts (GSE69914 [1.1], GSE225845 [1.2], GSE287331 [1.3]) and provide a unified, denoised, and biologically meaningful set of features to be used in the subsequent analyses of early epigenetic alterations.

1.1 Dataset GSE69914

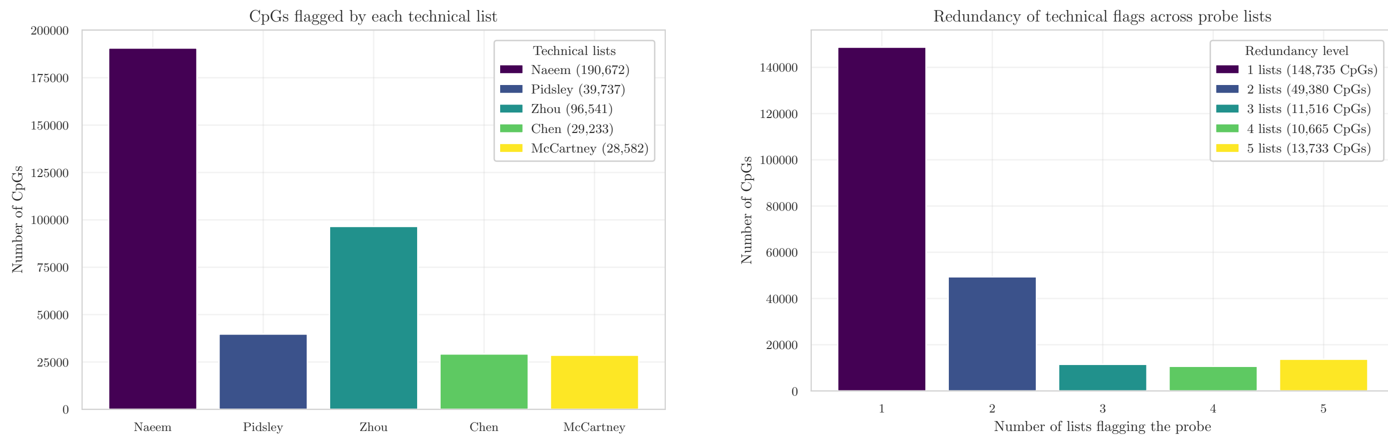
Step 0 – Initial Pre-processing The initial pre-processing stage verifies the structural integrity of the dataset. The β-value matrix and its corresponding phenotype table were imported from their Parquet representations. Both tables contained ✓ 397 samples, and the β-matrix included ✓ 485,512 CpGs, confirming full dataset availability at load time.

Sample identifiers in the phenotype table were inspected to ensure uniqueness and correct metadata structure. No duplicated `id_tissue` entries were detected ✓ (0 duplicates), and all required fields (`id_tissue`, `label`) were present ✓ (0 missing fields), indicating that the phenotype annotation is complete and coherent. Alignment between the phenotype table and the β-matrix was assessed by intersecting sample IDs. All samples were shared across matrices, yielding ✓ perfect correspondence (397/397 matched).

A probe-level and sample-level missingness check was then performed to verify the completeness of the methylation matrix. The entire dataset was found to be fully observed, with ✓ 0 missing CpG values and ✓ 0 missing sample entries. This confirms that no early NaN filtering or imputation is required for this dataset.

Step 1 – Infinium Type I/II Bias Diagnostics Raw methylation intensity data (*IDAT* files) were processed by the original authors using the `minfi` pipeline and `BMIQ` normalization, as reported in [1]. A well-known characteristic of Illumina Infinium microarrays is the **systematic technical bias between Type I and Type II probes**. This bias has been extensively documented in the literature, most notably by Maksimovic *et al.* [2], who showed that the two probe chemistries produce inherently different β-value distributions and therefore require appropriate probe-type normalization. Such differences, if uncorrected, can propagate into downstream statistical analyses.

To independently verify that this bias had been successfully mitigated in the present dataset, β-values were stratified by probe design (Type I vs. Type II) using the official Illumina 450K manifest from Bioconductor [3]. Following the diagnostic procedure of Teschendorff *et al.* [4], two complementary assessments were performed:



(a) CpGs flagged by each technical filtering resource (Naeem, Pidsley, Zhou, Chen, McCartney).

(b) Redundancy distribution of probes flagged by 1–5 lists.

Figure 2: Summary of technical filtering. **(a)** Intersection between the dataset and each external probe-filtering resource, showing the number of CpGs flagged by Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis indicating how many probes were simultaneously flagged by 1 to 5 independent resources, quantifying the concordance among technical annotations.

1. **Density comparison.** As shown in Figure 1a, the β-value distributions of Type I and Type II probes substantially overlap, indicating the absence of the multi-modal distortion typical of unnormalized data.

2. **Quantile–quantile alignment.** The Q–Q plot in Figure 1b exhibits quite a diagonal trend for Type II vs. Type I quantiles, which is the expected signature of balanced probe-type scaling after normalization.

The resulting pattern closely matches the characteristic behaviour described by Teschendorff *et al.* [4] and Wang *et al.* [5], while being fully consistent with the probe-type discrepancies originally analysed by Maksimovic *et al.* [2]. ✓ Together, these diagnostics confirm that the dataset has undergone effective Type I/II probe bias correction.

Step 2 – Technical Filtering Technical filtering aims to remove unreliable or biologically confounded probes prior to statistical modelling. This stage reduces noise, improves downstream reproducibility, and ensures that only high-confidence CpG loci are retained for analysis (Figure 2).

Exclusion of technical probe sets. A collection of curated probe resources was used to exclude loci known to exhibit technical artefacts (see Appendix A for details). These include probes affected by common genetic variants at the CpG or single-base extension site [6], [7], probes with demonstrated multi-mapping or off-target hybridisation [8], [9], and probes flagged by the `MASK_*` reliability annotations [6]. Additional quality control criteria follow the hierarchical filtering strategy of Naeem et al. [10], which targets multi-mapping loci, repeat structures, and disruptive SNPs.

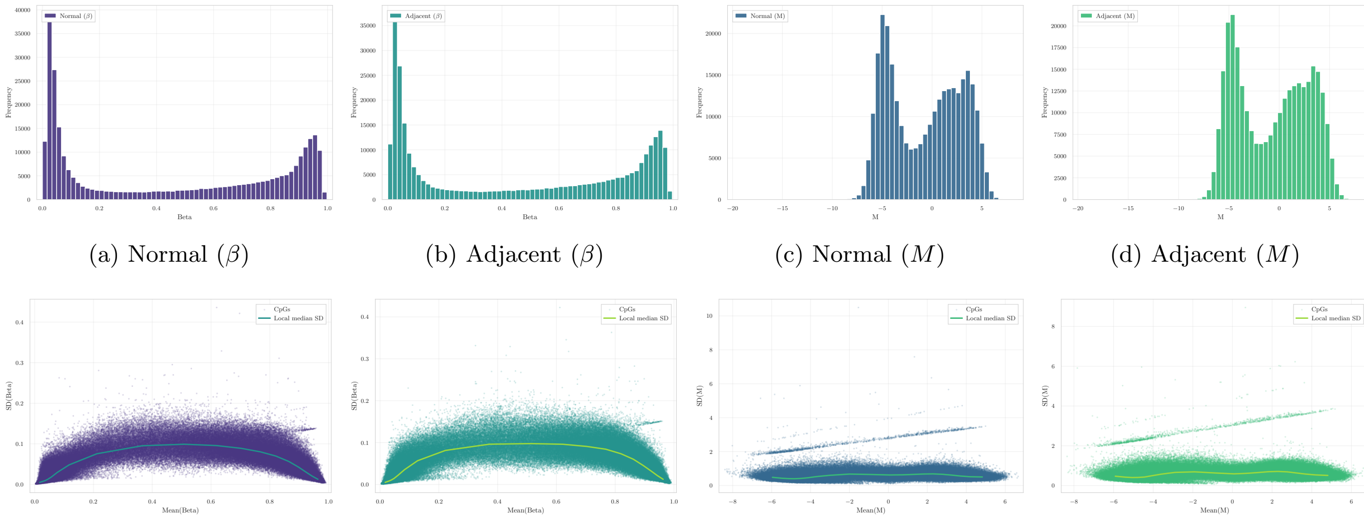
The overlaps between these resources and the β-matrix were substantial (Figure 2a). A total of ✓ 190,672 probes were flagged by Naeem et al., ✓ 39,737 by Pidsley et al., ✓ 29,233 by Chen et al., ✓ 96,541 by the Zhou `MASK_*` fields, and ✓ 28,582 by McCartney et al. These lists are known to partially overlap—particularly those focused on multi-mapping and SNP-associated unreliability—and their union resulted in the removal of ✓ 225,426 unique CpGs. After this operation, ✓ 260,086 CpGs remained.

Annotation-based filtering (Illumina manifest). To ensure consistency with the manufacturer's official annotation, the HumanMethylation450 v1.2 manifest [11] was used to validate probe identity and genomic mapping. Through this annotation, probes lacking valid chromosome information, those with undefined genomic coordinates, and those targeting non-CpG contexts (IDs beginning with "`ch`") were identified and removed. This step aligns the working feature set with Illumina's reference genome definition and follows the recommendations of Zhou et al. [6] and Pidsley et al. [7]. A total of ✓ 875 non-CpG probes were excluded, reducing the feature space to ✓ 259,211 CpGs.

Removal of sex-chromosome probes (X/Y). To avoid sex-related confounding effects and to maintain cross-dataset consistency, all probes located on chromosomes X and Y were removed based on manifest annotations. This choice is consistent with established preprocessing workflows for Illumina 450K data, where probes on sex chromosomes are excluded when male and female samples are mixed, as recommended by Maksimovic et al. [2]. Moreover, sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and joint normalization of autosomal and sex-chromosome probes has been shown to introduce artificial sex-related bias into thousands of autosomal CpGs [12]. Removing X/Y probes therefore provides a conservative and robust strategy when sex-specific methylation is not

Table 1: CpG filtering summary for the GSE69914 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final CpGs
485,512	190,672	39,737	96,541	29,233	28,582	875	7,728	251,483



(a) Normal (β) (b) Adjacent (β) (c) Normal (M) (d) Adjacent (M)

(e) Mean–SD Normal (β) (f) Mean–SD Adjacent (β) (g) Mean–SD Normal (M) (h) Mean–SD Adjacent (M)

Figure 3: **Distributional comparison between β and M quantifications in Normal and Normal-adjacent tissues.** Top row: β exhibits a bounded, U-shaped distribution, whereas M shows an approximately symmetric profile. Bottom row: The substantial mean-dependent variance in β is dramatically reduced after transformation, confirming that M-values provide a more homoscedastic scale for statistical modeling.

under investigation. This filtering step eliminated ✓ 7,728 CpGs, producing a final autosomal working set of ✓ 251,483 CpGs.

Overall, the initial set of ✓ 485,512 CpGs was reduced to ✓ 251,483 CpGs after applying all technical lists, manifest-based validation, and sex-chromosome exclusion. In total, ✓ 234,029 unique CpGs were removed across all criteria. A per-probe summary file `removed_cpgs_all_filters_summary_gse69914.csv` records, for each excluded probe, which resources contributed to its removal, ensuring full reproducibility of the filtering pipeline.

Step 3 – β to M Conversion Illumina Infinium methylation arrays measure methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) signal intensities per CpG. The processed dataset provides only β-values,

$$\beta = \frac{y^{(M)}}{y^{(M)} + y^{(U)}},$$

a bounded proportion in $[0, 1]$ characterised by strong mean–variance dependence. Following the foundational analysis of Du *et al.* [13], all β-values were transformed into M-values using the logit relation

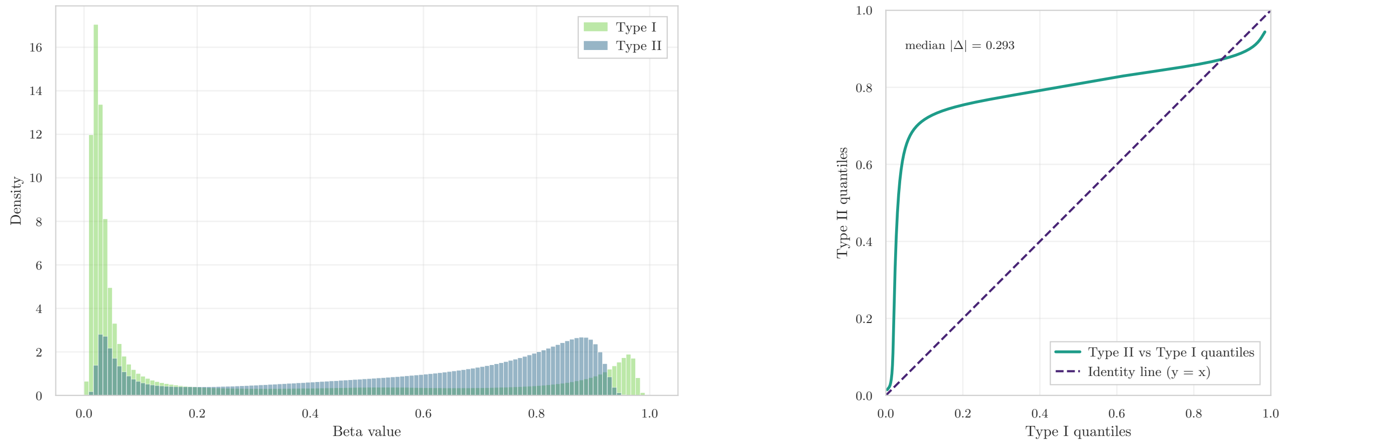
$$M = \log_2\left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon}\right), \qquad \varepsilon = 10^{-6},$$

which yields an approximately homoscedastic scale more suitable for linear modelling, empirical Bayes moderation, ComBat batch correction, and regression-based inference.

Importantly, the adoption of M-values is not only theoretically motivated but also widely implemented in practice. Several high-impact studies routinely perform statistical modelling in M-space while retaining β-values solely for biological interpretation and effect-size visualisation. Examples include the differential methylation analysis of purified human blood cell types by Reinius *et al.* [14], the methodological framework underlying the `minfi` pipeline described by Triche *et al.* [15], and oncological applications such as the tumour hypoxia study by Thienpont *et al.* [16], where all β-values were explicitly transformed to M-values prior to statistical testing. These examples illustrate that M-values have become a de facto standard for inferential analyses in Illumina 450K/EPIC workflows.

✓ The conversion was applied to the complete post-filtering matrix (251,483 CpGs across 397 samples).

Distributional consequences of the transformation. Figure 3 illustrates in detail how the $\beta \to M$ mapping reshapes the empirical distribution of methylation. Normal and Adjacent tissues display the expected U-shaped profiles in



(a) β-value distributions for Infinium Type I and Type II probes. The two probe families exhibit markedly different empirical shapes, indicating residual probe-design bias.

(b) Q–Q comparison of Type II vs. Type I β-value quantiles. The substantial deviation from the identity line ($y = x$), with median absolute deviation ✓ 0.2933, indicates incomplete probe-type alignment.

Figure 4: Diagnostic evaluation of Infinium Type I/II probe-design bias. Both the density distributions and the Q–Q pattern reveal that the two probe chemistries retain systematically different scaling behaviour, indicating that residual Type I/II bias persists in the dataset.

β-space, reflecting biological accumulation near fully unmethylated or fully methylated states. After logit transformation, the corresponding M-value distributions become substantially more symmetric and comparable across groups, indicating that the transformation regularises the scale without distorting biological structure.

The mean–SD relationships further highlight this effect. In β-space, variance is strongly mean-dependent and inflates near the boundaries, whereas M-values exhibit markedly flatter SD profiles. This variance-stabilising behaviour is consistent with the theoretical findings of Du *et al.* [13] and with the practical motivations underlying the adoption of M-values in modern methylation studies.

1.2 Dataset GSE225845

Step 0 – Initial Pre-processing The β-value matrix was successfully imported from its Parquet representation, yielding ✓ 477 samples and ✓ 750,426 CpGs, reflecting the high dimensionality of the Illumina EPIC platform. The corresponding phenotype table contained the same number of samples ✓ (477 entries) and provided all required metadata fields with no inconsistencies.

Sample identifiers were inspected to ensure uniqueness and correct formatting. No duplicated `id_tissue` values were detected, and all metadata columns required for downstream analysis were present ✓ (0 missing fields). Alignment between phenotype records and the methylation matrix was confirmed by intersecting sample IDs, producing ✓ perfect correspondence (477/477 matched).

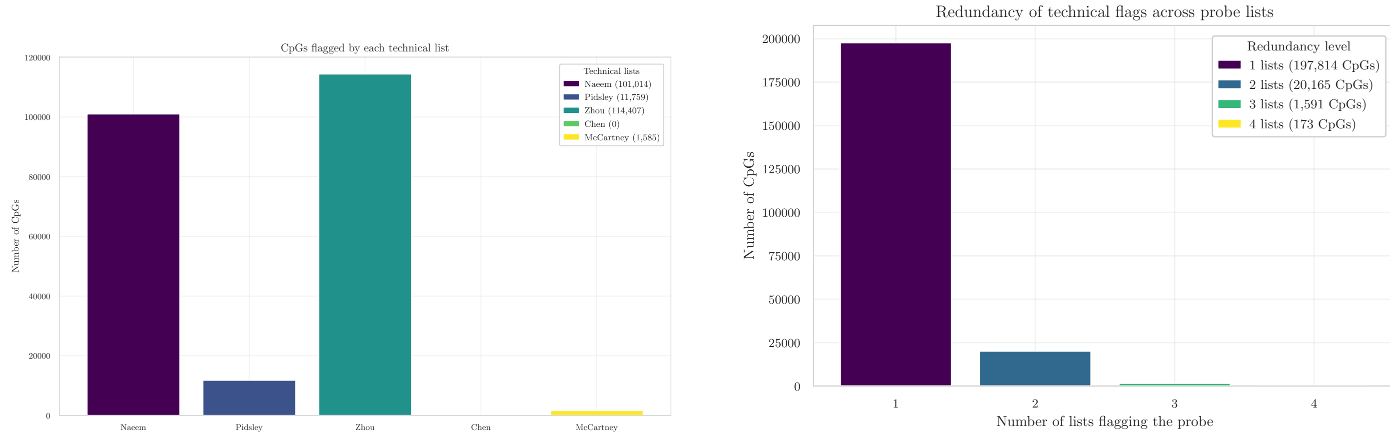
A probe-level and sample-level missingness check verified the completeness of the methylation data. The dataset was fully observed, with ✓ 0 missing CpG values and ✓ 0 missing sample entries. Accordingly, no early NaN filtering or imputation was required before proceeding to probe-type diagnostics and technical filtering.

Step 1 – Infinium Type I/II Bias Diagnostics All probes were stratified, this yielded ✓ 119,760 Type I and ✓ 627,842 Type II probes.

The density representation of the two probe families (Figure 4a) reveals a marked discrepancy between their empirical β-value distributions. Type I probes exhibit a highly concentrated pattern near the unmethylated and fully methylated boundaries, whereas Type II probes display a broader and right-shifted distribution. Such divergence indicates that the two probe chemistries retain systematically different scaling behaviour.

This finding is reinforced by the quantile–quantile comparison in Figure 4b. Type II quantiles deviate substantially from the identity line, with a ✓ median absolute deviation of 0.2933, demonstrating that the expected alignment between the two probe types is not present. The curvature of the Q–Q profile confirms the presence of a residual probe-design imbalance that has not been fully corrected by preprocessing.

Step 2 – Technical Filtering Beginning from an initial set of ✓ 750,426 CpGs, the intersection with the external probe-filter lists showed substantial overlap across multiple sources.



(a) Number of CpGs flagged by each technical list.

(b) Redundancy distribution: CpGs flagged by 1–4 lists.

Figure 5: Summary of technical filtering for the methylation dataset. **(a)** Overlap between CpGs and the Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis quantifying concordance among the technical annotations.

Table 2: CpG filtering summary for the GSE225845 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	ChenX	McCartney	Non-CpG	X/Y	Final CpGs
750,426	101,014	11,759	114,407	0	1,585	773	14,071	530,683

Exclusion of technical probe sets. The curated technical resources flagged a substantial fraction of probes as potentially unreliable (Figure 2a): ✓ 101,014 CpGs were flagged by Naeem, ✓ 11,759 by Pidsley, ✓ 114,407 by the Zhou fields, ✓ 0 by Chen, and ✓ 1,585 by McCartney. The union of all technical lists resulted in the removal of ✓ 204,899 unique CpGs, leaving ✓ 545,527 CpGs for subsequent steps.

Annotation-based filtering (Illumina manifest). Using the EPIC v1.0 manifest, all probes lacking valid genomic coordinates or targeting non-CpG contexts were removed. This step eliminated ✓ 773 non-CpG loci, reducing the working set to ✓ 544,754 CpGs.

Removal of sex-chromosome probes (X/Y). Probes mapped to chromosomes X and Y were excluded. A total of ✓ 14,071 CpGs were removed, producing a filtered autosomal set of ✓ 530,683 CpGs.

Across all steps, ✓ 219,743 unique CpGs were removed from the initial ✓ 750,426, yielding a final feature set of ✓ 530,683 CpGs. A detailed per-probe summary of the filtering decisions is provided in `removed_cpgs_all_filters_summary_gse225845.csv`.

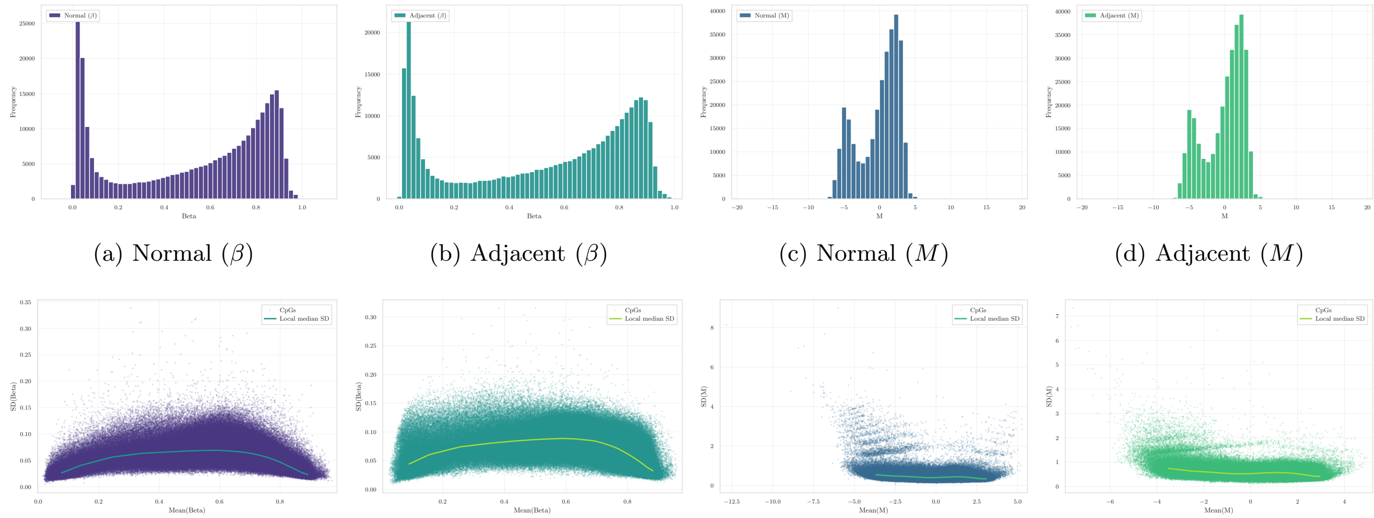
Step 3 – β to M Conversion All β-values were transformed into M-values using the standard logit relation yielding an unbounded and approximately homoscedastic scale suitable for linear modelling and batch correction procedures. The transformed dataset, denoted `M_clean`, preserves the same sample structure and CpG ordering as the β matrix and was saved in Parquet format for downstream analysis.

Distributional effects of the transformation. The empirical distributions of Normal and Adjacent samples exhibit the characteristic U-shape in β-space and the expected approximately symmetric profiles after conversion. The mean–SD relationship demonstrates the well-known variance inflation at intermediate β levels, which becomes substantially stabilised in M-space (Figure 6). These patterns confirm that the M-value representation provides a more appropriate scale for statistical inference.

The filtered β-matrix (✓ 530,683 CpGs, ✓ 477 samples) was successfully converted into the corresponding M-value representation and stored as a Parquet object.

1.3 Dataset GSE287331

Step 0 – Initial Pre-processing The dataset was imported from its Parquet representation, yielding an initial β-value matrix of ✓ 446 samples and ✓ 866,552 CpGs. The accompanying phenotype table reported the same number of entries and contained all required metadata fields. Before continuing with probe-level preprocessing, the dataset underwent a dedicated missingness-cleaning stage to ensure high-quality input for downstream analysis. After harmonising phenotype labels and applying dataset-specific selection rules, only samples belonging to the three core tissue states of interest—Normal, Adjacent, and Tumour—were kept. This resulted in a final working cohort of ✓ 311 samples, used for all downstream preprocessing and analysis.



(a) Normal (β) (b) Adjacent (β) (c) Normal (M) (d) Adjacent (M)

(e) Mean–SD Normal (β) (f) Mean–SD Adjacent (β) (g) Mean–SD Normal (M) (h) Mean–SD Adjacent (M)

Figure 6: **Distributional and variance-structure comparison of β and M values in Normal and Adjacent breast tissue.** Top row: β- and M-value histograms, showing the characteristic U-shape in β-space and the approximate symmetry induced by the logit transformation. Bottom row: mean–SD relationships illustrating how the strong mean-dependent variance in β-space is markedly stabilised in M-space.

Missingness filtering and imputation. Probe-wise missingness was quantified across the β-matrix. Consistent with recommendations from Islam *et al.* [17] and Mansell *et al.* [18], a stringent threshold was applied: all CpG sites missing in more than ✓ 1% of samples were removed. This filter ensures high-confidence measurements while leveraging the large dimensionality of the EPIC platform.

After applying this criterion, the remaining level of sparsity was extremely low, with only ✓ 58,561 NaN values (✓ 0.0187% of all entries). These residual missing values were imputed using probe-wise median substitution, a strategy that avoids introducing group-specific bias and is also adopted in recent harmonisation workflows such as mLiftOver [19].

✓ The resulting β-matrix contained no missing values.

Data integrity verification. After missingness filtering and imputation, the final working matrix consisted of ✓ 311 samples and ✓ 702,916 CpGs. All entries were fully observed (✓ 0 residual NaN).

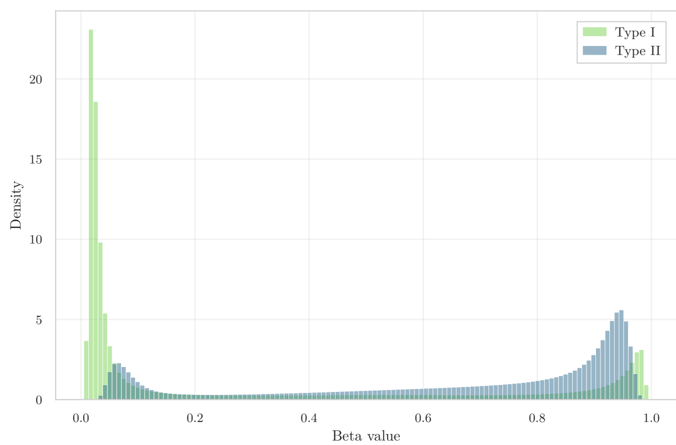
Step 1 – Infinium Type I/II Bias Diagnostics Using the official EPIC v1.0 manifest, ✓ 115,705 Type I and ✓ 585,514 Type II probes were identified. The β-value distributions of the two chemistries showed a marked mismatch (Figure 7a), with Type II probes displaying a strong right-shifted profile. The quantile–quantile comparison confirmed a substantial systematic offset, with a median absolute deviation of ✓ 0.380 (Figure 7b), indicating the absence of any Infinium probe-design bias correction in the originally released data.

Given this discrepancy, a per-sample BMIQ normalization step was applied using the `wateRmelon` implementation, operating strictly on individual samples to minimise cross-sample distortion. Post-correction diagnostics showed a modest reduction in probe-type mismatch ($|\Delta| : 0.380 \rightarrow 0.370$) and a slight improvement in the alignment of high-β regions, but the overall distributional discrepancy remained substantial. Due to this limited harmonization and the known tendency of BMIQ to attenuate biological differences when global distributional shifts are present, the corrected matrix is not used for downstream modelling. All subsequent analyses therefore rely on the non-BMIQ version of the dataset, which preserves clearer group-level structure and avoids normalization-induced shrinkage (for more details, see *01 – Report Data Exploration & Visualization*.).

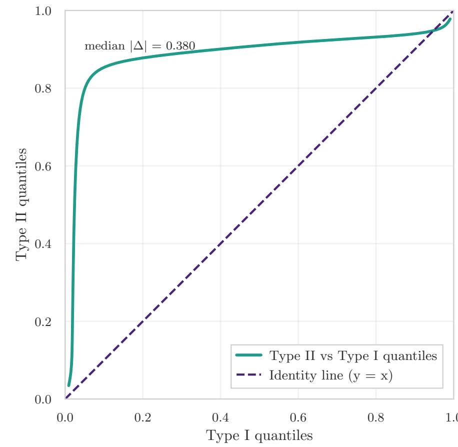
Step 2 – Technical Filtering Technical probe filtering excluded CpG loci affected by unreliable hybridisation, SNP interference, multi-mapping or annotation inconsistencies. Across the established probe-filtering resources, ✓ 124,511 CpGs were flagged by Naeem, ✓ 8,992 by the Pidsley blacklist, ✓ 102,802 by Zhou's fields, ✓ 1,730 by Chen's cross-reactive probe set, and ✓ 3,672 by McCartney's cross-hybridisation catalogue. The union of these lists removed ✓ 203,114 unique CpGs, leaving ✓ 499,804 CpGs prior to annotation-based filtering.

Annotation-based filtering (Illumina manifest). Using the EPIC v1.0 manifest, probes lacking valid genomic coordinates or targeting non-CpG contexts were removed. This eliminated ✓ 500 non-CpG probes, reducing the feature set to ✓ 499,304 CpGs.

Removal of sex-chromosome probes (X/Y). Probes mapped to chromosomes X and Y were removed to avoid

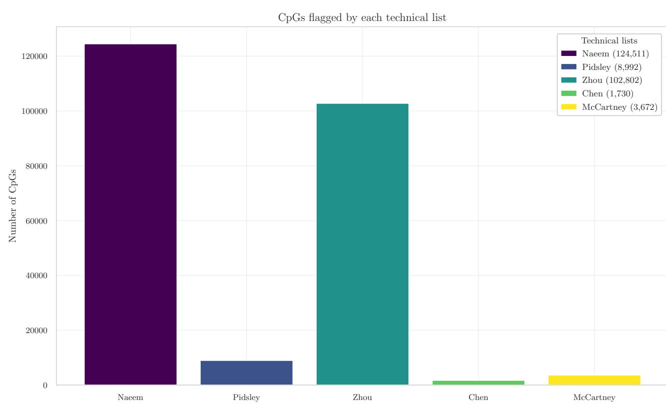


(a) Type I vs. Type II density (pre BMIQ).

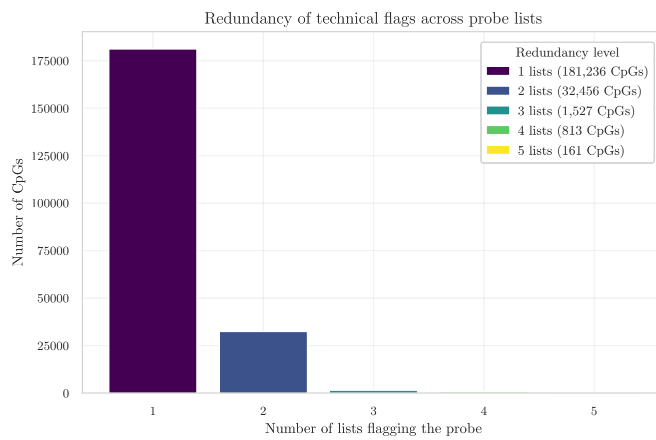


(b) Q–Q comparison of Type II vs. Type I probes (pre BMIQ). The median absolute deviation of ✓ 0.380 indicates a strong quantile-wise shift, confirming the absence of probe-type normalization in the original dataset.

Figure 7: Diagnostic evaluation of Infinium probe–design bias before and after BMIQ. The correction reduces the discrepancy only marginally, and substantial distributional differences remain between probe types.



(a) CpGs flagged by each technical list.



(b) Redundancy of probes flagged by 1–5 lists.

Figure 8: Summary of technical filtering for the methylation dataset. **(a)** Number of CpGs intersecting each curated resource. **(b)** Redundancy analysis capturing concordance across filtering lists.

sex-driven confounding. A total of ✓ 12,579 CpGs were discarded, resulting in an autosomal working set of ✓ 486,725 CpGs.

Across all filters, ✓ 216,193 unique CpGs were removed from the initial ✓ 866,552, leaving a final autosomal set of ✓ 486,725 CpGs. A probe-level removal summary is provided in `removed_cpgs_all_filters_summary_gse287331.csv`.

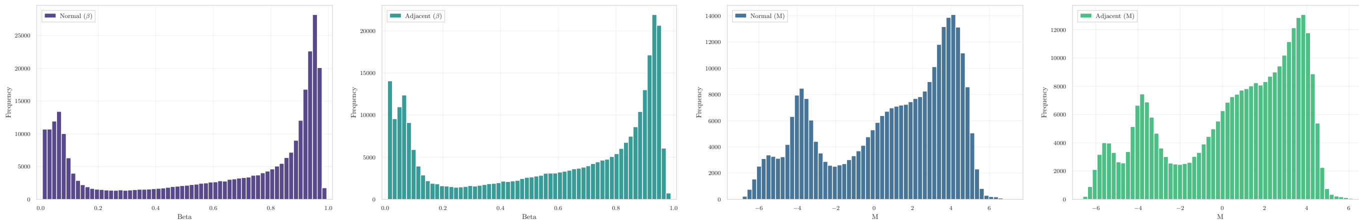
Step 3 – β to M Conversion Given the strong mean–variance dependence intrinsic to β-values, the working matrix obtained after technical and invariance filtering (✓ 486,725 CpGs across ✓ 311 samples) was transformed into M-values. This transformation yields an approximately homoscedastic scale and is the standard input for linear models, empirical Bayes moderation, and batch-effect correction.

The transformation was applied to all CpG columns (✓ 284,745 CpGs, excluding `id_tissue` and `label`). The resulting object, `M_clean`, preserves the same schema as the β-matrix and was stored as a memory-efficient `LazyFrame`.

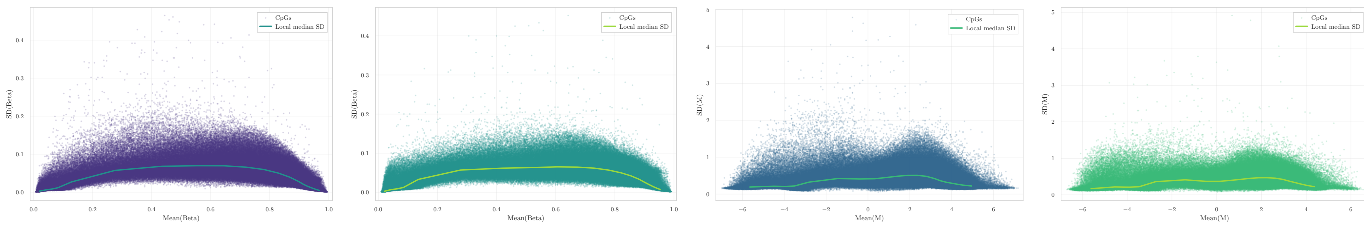
Distributional effects of the $\beta \to M$ transformation. Figure 9 illustrates the impact of the transformation on the empirical data distribution. As expected, the U-shaped β-profiles for Normal and Adjacent tissues become substantially more symmetric after conversion, while the pronounced mean–SD dependence flattens into a markedly more stable variance structure across the full range of M-values. These properties make M-values preferable for statistical modelling, whereas β-values are retained for biological interpretation and visualisation.

Table 3: CpG filtering summary for the GSE287331 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final CpGs
866,552	124,511	8,992	102,802	1,730	3,672	500	12,579	486,725



(a) Normal (β) (b) Adjacent (β) (c) Normal (M) (d) Adjacent (M)



(e) Mean–SD (Normal, β) (f) Mean–SD (Adjacent, β) (g) Mean–SD (Normal, M) (h) Mean–SD (Adjacent, M)

Figure 9: **Distributional and variance-structure comparison of β and M values in Normal and Adjacent breast tissue.** Top row: empirical β- and M-value distributions, illustrating the U-shaped behaviour of β-values and the symmetry introduced by the logit transformation. Bottom row: mean–SD relationships showing strong mean-dependent variance in β-space and its stabilisation in M-space, supporting the use of M-values for downstream modelling.

2 Inter-Dataset Analysis

After completing the intra-dataset pre-processing for each cohort, a concise inter-dataset analysis is performed to summarise the effects of the pipeline and to motivate the subsequent feature selection step.

Pverview of the pre-processing pipeline All three datasets underwent a harmonised pre-processing workflow, including initial integrity checks, literature-based technical filtering, manifest-driven probe validation, removal of sex-chromosome loci, and transformation from β-values to M-values. While the specific implementation details differ slightly depending on platform characteristics and data quality, the conceptual structure of the pipeline is consistent across datasets.

Table 4 reports, for each dataset, the number of samples retained and the dimensionality of the methylation matrix before and after pre-processing. Despite substantial differences in initial platform coverage (450K vs EPIC), technical filtering consistently removes a large fraction of probes, yielding cleaner and more reliable feature spaces.

CpG overlap across datasets To further characterise the relationship between datasets, CpG overlap was evaluated both before and after pre-processing. Figure 10a illustrates the overlap of CpG identifiers across the three datasets prior to filtering, reflecting platform design constraints and baseline annotation differences. As expected, the intersection is dominated by probes shared between the two EPIC datasets, with a smaller but substantial core common to all three cohorts.

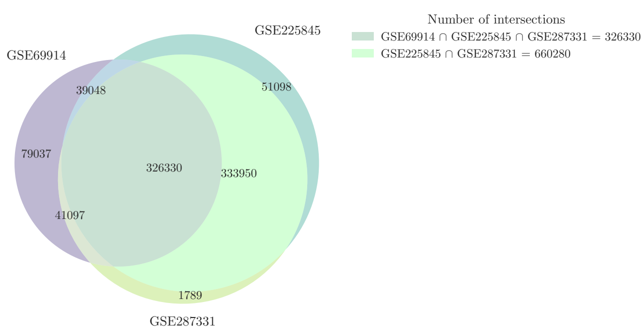
Figure 10b reports the corresponding overlap after the complete pre-processing pipeline. Although the total number of CpGs is markedly reduced, a large shared core remains, indicating that technical filtering primarily removes platform- or annotation-specific probes while preserving a common, high-confidence CpG backbone.

Rationale for feature selection Despite extensive technical filtering, the post-processed datasets remain highly dimensional, with hundreds of thousands of CpGs retained in each cohort. Moreover, the large shared CpG intersection across datasets suggests that technical artefacts have been effectively mitigated, shifting the primary challenge from data quality to signal extraction.

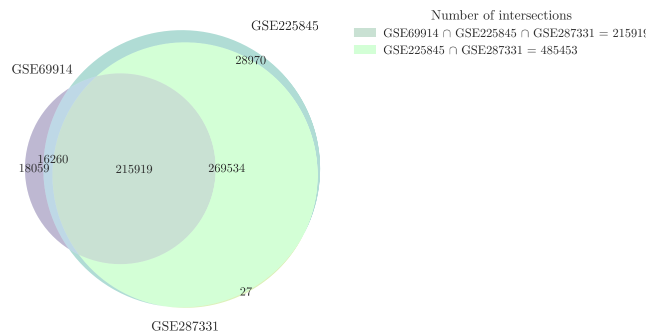
At this stage, the limiting factor is no longer technical reliability but statistical and biological relevance. The remaining feature space is expected to contain a substantial proportion of CpGs that are either weakly informative or irrelevant

Table 4: Summary of sample size and CpG dimensionality before and after pre-processing.

Dataset	Platform	Normal	Adjacent	CpGs *before* pre-processing	CpGs *after* pre-processing
GSE69914	450K	50	42	485,512	251,483
GSE225845	EPIC	113	140	750,426	530,683
GSE287331	EPIC	182	60	866,554	486,725



(a) CpG overlap across datasets before pre-processing.



(b) CpG overlap across datasets after pre-processing.

Figure 10: Comparison of CpG overlap across datasets before and after pre-processing.

for distinguishing Normal and Adjacent breast tissue. Consequently, a dedicated feature selection strategy is required to identify a reduced subset of CpGs that captures biologically meaningful variation while improving interpretability, robustness, and downstream modelling performance.

The next chapter therefore focuses on feature selection methodologies applied to these pre-processed datasets, building on the stable and harmonised CpG space established here.

A Filtering lists

This appendix details the major technical filtering lists and annotations applied in Illumina 450K and EPIC methylation arrays to remove unreliable or ambiguous CpG probes.

A.1 Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the humanmethylation450 array

Naeem *et al.* [10] proposed a structured filtering framework to reduce false discoveries by sequentially discarding probes based on hybridisation specificity and genomic integrity (Figure 11). The main exclusion steps are:

- Multi-mapping or cross-hybridising probes ⇒ *discard.*
- Probes overlapping repetitive elements (LINE/SINE/ALU) ⇒ *discard.*
- Probes targeting regions with INDELs ⇒ *discard.*
- Probes overlapping SNPs:
  - SNP at CpG or extension site ⇒ *discard.*
  - SNP near target but not interfering in bisulfite space ⇒ *keep.*

The resulting "discard" set therefore removes probes affected by cross-reactivity, structural polymorphisms (INDELs), and SNPs that alter CpG interrogation.

A.2 Discovery of cross-reactive probes and polymorphic cpgs in the illumina infinium humanmethylation450 microarray

Chen *et al.* [8] empirically identified probes in the 450K array that exhibit off-target hybridisation or overlap with common SNPs. For this project, only the **cross-reactive list** is available and used. This list enumerates ∼29k multi-mapping probes whose methylation signal cannot be uniquely attributed to one genomic locus.

Table 5: Technical categories covered by each filtering resource.

Category	Naeem A.1	Chen A.2	Pidsley A.3	Zhou A.4	McCartney A.5
Cross-hybridisation / multi-mapping	Yes (Steps 1–2)	Yes	Yes	`MASK_mapping`	Table 2+3
SNP at CpG / extension base	Yes (Step 5)	–	Yes	`MASK_snp5, MASK_extBase`	–
Nearby SNP tolerated	Conditional (Step 6)	–	–	–	–
INDEL / structural variant	Yes (Step 3)	–	–	–	–
Non-CpG probes	–	–	Yes	`MASK_nonCG`	Table 3

A.3 Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling

Pidsley *et al.* [7] provide a platform-level assessment of the EPIC array, with explicit *annotated probe lists* that flag probes whose methylation signal may be confounded by design- or genome-related artefacts. The focus is on probe categories and positions where technical bias arises, rather than on a universal, prescriptive blacklist. In particular, they catalogue:

- **Cross-hybridising CpG-targeting probes:** CpG probes showing sequence homology (off-target matches) to additional genomic loci, yielding non-unique hybridisation and potentially inflated or ambiguous β signals.
- **Cross-hybridising non-CpG-targeting probes:** off-target issues among CNG/non-CpG probes; these are typically excluded in CpG-centric analyses due to limited interpretability and higher risk of artefacts.
- **Probes overlapping common genetic variation:** annotation of variants from population data at three critical positions:
  - *At the interrogated CpG* (polymorphic CpG) — directly affects the presence of the CpG dinucleotide and the measured methylation state;
  - *At the single-base extension (SBE) site* (Type I) — perturbs extension and dye chemistry, biasing intensity ratios;
  - *Within the probe body* — can reduce binding affinity or alter hybridisation kinetics, especially for common variants.

A.4 Comprehensive characterization, annotation and innovative use of infinium dna methylation beadchip probes

Zhou *et al.* [6] released a unified probe annotation for both 450K and EPIC arrays with multiple binary mask columns (`MASK_*`). Each mask flags probes to be removed due to specific technical artifacts.

- `MASK.mapping`: probes with low or inconsistent mapping quality (non-unique alignment or presence of INDELs);
- `MASK.typeINextBaseSwitch`: Type-I probes carrying a SNP in the extension base that causes a color-channel switch (*CCS* probes);
- `MASK.extBase`: probes whose extension base is inconsistent with the expected color channel or CpG context;
- `MASK.sub30.copy`: probes with non-unique 30-bp 3' subsequences (potential cross-hybridisation);
- `MASK.snp5.common`: probes overlapping a SNP within ±5 bp of the interrogated CpG (even with global MAF ¡ 1%);
- `MASK.snp5.GMAF1p`: probes overlapping SNPs with global MAF ¿ 1%;
- `MASK.general`: recommended composite mask integrating mapping, SNP, and cross-reactivity filters for general use;
- `MASK.rmsk15`: probes overlapping RepeatMasker regions (not recommended for exclusion in standard workflows).

This resource provides a programmatic and reproducible way to perform fine-grained probe masking.

A.5 Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip

McCartney *et al.* [9] assessed probe design artifacts and published supplementary tables that include:

- Cross-hybridising CpG-targeting probes;
- Cross-hybridising non-CpG-targeting probes.

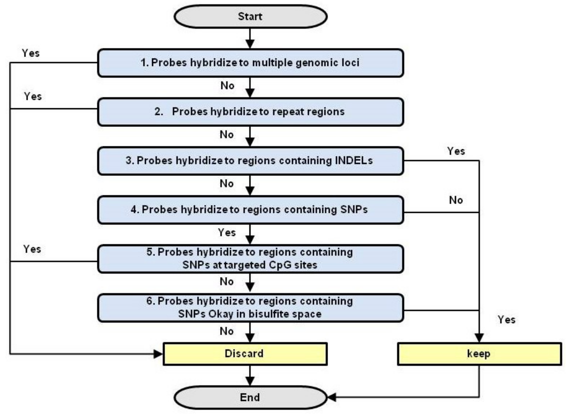


Figure 11: Workflow for determining affected Probes.

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